

Cohort Retention Basics

Project Report - Data Analytics Track

Veda Technology Internship

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Tools: Python (pandas, matplotlib, seaborn) + SQL (PostgreSQL)

Dataset: Online Retail II (UCI)

1. Objective

Build a cohort retention table grouped by signup month and understand how customer retention changes over time. Deliverables: cohort table, heatmap and insights.

2. Dataset

Online Retail II contains transactions of a UK-based online gift retailer between 1 Dec 2010 and 9 Dec 2011 (541,910 rows, 8 columns: Invoice, StockCode, Description, Quantity, InvoiceDate, Price, Customer ID, Country). There is no explicit 'signup date', so a customer's **first purchase month** is used as the signup month.

3. Method

- **Cohort** = the month in which a customer made their first purchase.
- **Cohort index** = number of months between a purchase and the customer's first purchase (0 = signup month).
- **Retention window** = calendar months (consistent for every cohort).
- **Retention %** = unique customers active in month N / cohort size (month 0) x 100.
- The same logic was built twice: in Python (pandas) and in SQL (PostgreSQL views). Both produce identical counts.

4. Data Cleaning

Step	Rows affected
Raw rows	541,910
Missing Customer ID (removed)	135,080
Cancelled invoices - Invoice starts with 'C' (removed)	8,905
Dec-2011 rows (incomplete month, data ends 9 Dec) (removed)	excluded
Clean rows used	380,580
Unique customers	4,297
Cohorts (Dec-2010 to Nov-2011)	12

December 2011 was excluded because only 9 days of data exist; including it would make the last diagonal of the table look like a fake drop in retention.

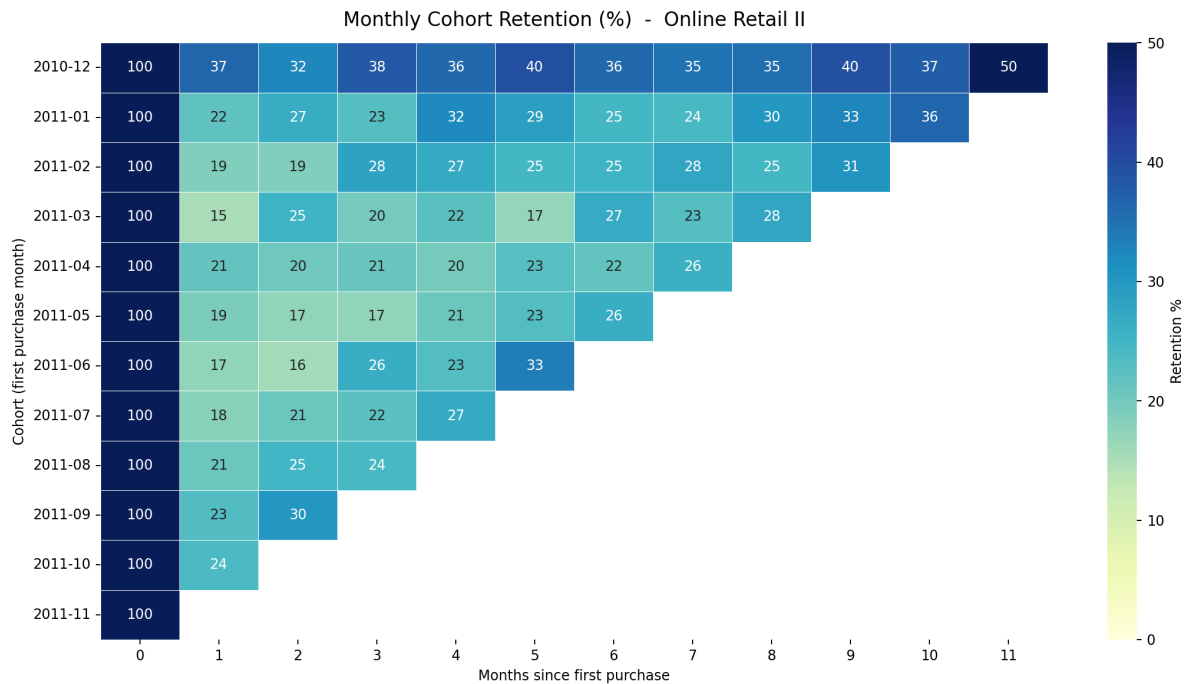
5. Cohort Table (active customers)

[illegible]

Cohort Retention Table (%)

[illegible]

6. Heatmap



Darker cells = higher retention. Blank cells are months that have not happened yet for that cohort.

7. SQL Approach (PostgreSQL)

- **retail_raw** - table created and CSV imported through pgAdmin (encoding LATIN1).
- **retail_clean** - view: removes NULL customers, cancellations, bad rows, converts text date with `to_timestamp()`.
- **customer_cohort** - view: `MIN(order_month)` per customer = cohort month.
- **cohort_activity** - view: `distinct (customer, month)` with `cohort_index` from `EXTRACT(YEAR/MONTH)`.
- **cohort_counts / cohort_retention** - views: `COUNT(DISTINCT customer_id)` and retention %.
- Final query pivots the long table into the wide cohort table using `MAX(CASE WHEN cohort_index = N ...)`.

8. Key Insights

- **Most customers do not come back quickly.** On average only about 24% of a cohort buys again in month 1 (size-weighted across 11 cohorts); month 3 is about 27% and month 6 about 29%.
- **The December 2010 cohort is the strongest.** It is also the largest (885 customers) and keeps 32-40% retention in almost every month, reaching 50.3% in Nov 2011. These are likely loyal or wholesale-type buyers.
- **2011 cohorts are weaker at the start.** Month-1 retention for Jan-Oct 2011 cohorts ranges from about 15% (March) to 24% (October); March 2011 is the weakest cohort despite having 452 new customers.
- **Retention rises later in the year.** Many cohorts show a lift in months 8-10 (Sep-Nov 2011), which matches the pre-Christmas buying season for a gift retailer, so seasonality affects retention numbers.
- **Autumn cohorts (Sep-Oct 2011) start better** (23-24% in month 1) than spring/summer cohorts (15-21%), so customers acquired near the holiday season are more likely to return quickly.
- **Cohort sizes:** acquisition dropped from 452 (Mar) to 169 (Aug) and recovered to 358 (Oct), so acquisition is seasonal too.

Recommendations

- Run a win-back or second-purchase offer within 30 days of the first order, since the biggest drop happens right after month 0.
- Study what the Dec-2010 customers bought and how they were acquired, and repeat it.
- Prepare retention campaigns before the Sep-Nov peak season.

9. Limitations

- Signup month is approximated by first purchase month; customers who bought before Dec 2010 appear as 'new' in Dec 2010, which inflates that cohort.
- Averages for later months (M9-M11) rely on very few cohorts and should be read carefully.
- Only 13 months of data; year-on-year seasonality cannot be separated from real retention change.

10. Interview Questions

What is a cohort?

A cohort is a group of users who share a common starting event in the same time period, for example customers whose first purchase was in January 2011. Cohorts let us compare groups fairly over the same lifetime.

Why is cohort retention useful?

Overall retention hides trends because new and old customers are mixed. Cohort retention shows whether newer customers stay longer or leave faster than older ones, which tells us if product, marketing or seasonality changes are working, and helps estimate customer lifetime value.

11. Conclusion

The cohort analysis shows a typical e-commerce pattern: a sharp drop after the first month, then a flat or slowly rising tail. The Dec-2010 cohort retains far better than later ones, and end-of-year seasonality lifts retention. The same result was reproduced in Python and SQL, which validates the pipeline.